

Lecture 4

Dynamics of rigid blades

ASDA / Dynamics of Rotor Systems

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QB 2.64 UoB

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This lecture

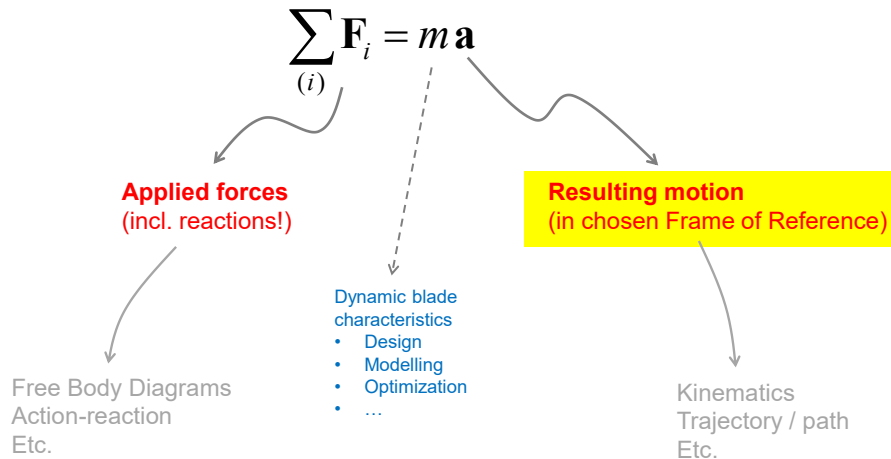
- Part 1: Elements of rigid body dynamics
- Part 1: Rigid blade model
- Part 2: Blade flapping motion
- Part 2: Blade lead-lag motion
- Part 3: Blade natural frequencies

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Newton's method recall

We will use Newton's method. It transforms dynamics to static problems and introduces the concept of **dynamic equilibrium**.



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Revise basic 2D kinematics

Motion is described: position $\mathbf{r}$, velocity $\mathbf{v}$, acceleration $\mathbf{a}$

Position vector: $\mathbf{r} = \mathbf{r}(t)$

Velocity vector: $\dot{\mathbf{r}} = \mathbf{v} = \boldsymbol{\Omega} \times \mathbf{r}$

Acceleration vector:

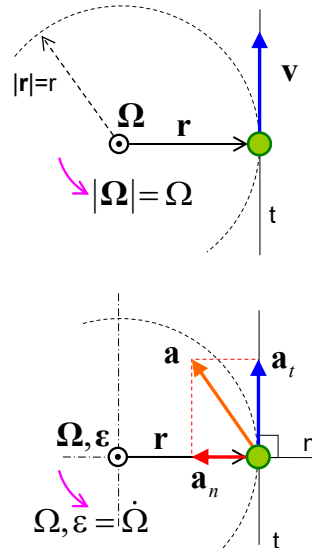
$$\ddot{\mathbf{r}} = \dot{\mathbf{v}} = \dot{\boldsymbol{\Omega}} \times \mathbf{r} + \boldsymbol{\Omega} \times \dot{\mathbf{r}} = \boldsymbol{\varepsilon} \times \mathbf{r} + \boldsymbol{\Omega} \times \mathbf{v}$$

$$\mathbf{a} = \boldsymbol{\varepsilon} \times \mathbf{r} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}) = \mathbf{a}_t + \mathbf{a}_n$$

tangential
acceleration

normal / centripetal
acceleration

$$a_n = |\mathbf{a}_n| = r \Omega^2$$

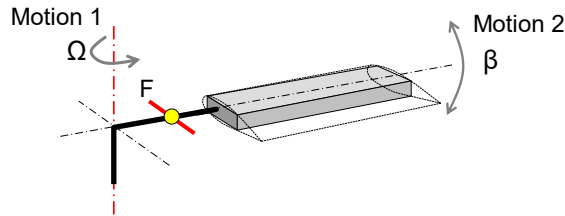


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Complex (or combined) motions

Consider rigid **flapping** and **rotating** blade, i.e. two constituent motions, where rotation is *prescribed* motion and flapping is *free* motion.



We want to predict (e.g. simulate) flapping. However, flapping motion alone does not happen in the *inertial frame of reference* (F hinge moves). This situation requires special approach leading to the *modified* Newton's method.



... denotes supporting material, not needed for exam.

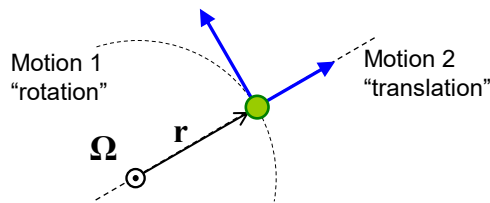
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Kinematics of combined motions

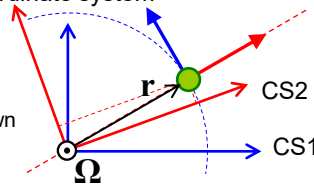
Point rotating in 2D can be seen as a mass rotating on an infinitely axially stiff string. Consider the case with a *soft* string (e.g. *axial vibrations of rotating blades*).



... break down the complicated motion to a sequence of simple constituent motions, each motion is linked with one coordinate system

Coordinate systems (CS):

- Motion 1 in CS1 (blue): relationship CS1-CS2 known
- Motion 2 in CS2 (red): motion of interest



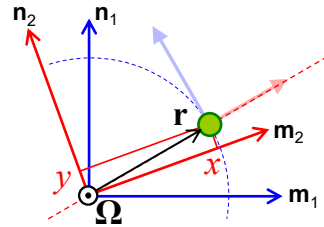
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Kinematics of combined motions

Any coordinate system is characterized by its *base vectors* (vectors $\mathbf{m}$ and $\mathbf{n}$).



Notes:

- base vectors have length 1
- base vectors are orthogonal
- CS1 vectors are stationary
- CS2 vectors rotate

We write $\mathbf{r}$ using base vectors in CS2

$$\mathbf{r} = x \mathbf{m}_2 + y \mathbf{n}_2 \quad \dots \text{note that both the coordinates and the base vectors change with time!}$$

$$\dot{\mathbf{r}} = \dot{x} \mathbf{m}_2 + x \dot{\mathbf{m}}_2 + \dot{y} \mathbf{n}_2 + y \dot{\mathbf{n}}_2 \quad \dots \text{first and third term is velocity in CS2 } \mathbf{v}_2 = \dot{x} \mathbf{m}_2 + \dot{y} \mathbf{n}_2$$

$$\dot{\mathbf{r}} = \mathbf{v}_2 + x(\boldsymbol{\Omega} \times \mathbf{m}_2) + y(\boldsymbol{\Omega} \times \mathbf{n}_2) \quad \dots \mathbf{m}_2 \text{ and } \mathbf{n}_2 \text{ derivatives use (in CS1) } \dot{\mathbf{e}} = \boldsymbol{\Omega} \times \mathbf{e}$$

$$\dot{\mathbf{r}} = \mathbf{v}_2 + \boldsymbol{\Omega} \times (x \mathbf{m}_2 + y \mathbf{n}_2)$$

$$\dot{\mathbf{r}} = \mathbf{v}_2 + \boldsymbol{\Omega} \times \mathbf{r} \quad \text{Total velocity in CS1} = \text{relative velocity CS2} + \text{velocity due to CS2-CS1!}$$

Very important formula for rotating systems!

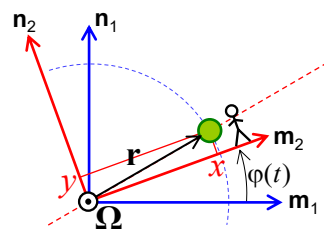
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Kinematics of combined motions

Absolute acceleration:



To use the Newton's method we need absolute acceleration of a material point: $\ddot{\mathbf{r}} = ?$

... note that $\mathbf{v}_2$ and $\mathbf{r}$ are vectors defined in CS2!

$$\ddot{\mathbf{r}} = \dot{\mathbf{v}}_2 + (\boldsymbol{\Omega} \times \mathbf{r})'$$

$$\ddot{\mathbf{r}} = \mathbf{a}_2 + \boldsymbol{\Omega} \times \mathbf{v}_2 + \dot{\boldsymbol{\Omega}} \times \mathbf{r} + \boldsymbol{\Omega} \times (\mathbf{v}_2 + \boldsymbol{\Omega} \times \mathbf{r})$$

differentiate $\mathbf{v}_2$ in CS1 differentiate $(\boldsymbol{\Omega} \times \mathbf{r})$ in CS1

$$\ddot{\mathbf{r}} = \mathbf{a}_2 + \boldsymbol{\epsilon} \times \mathbf{r} + 2\boldsymbol{\Omega} \times \mathbf{v}_2 + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r})$$

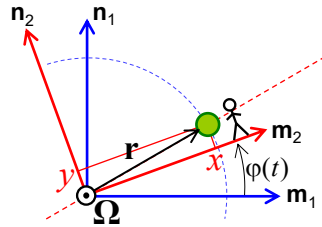
$\mathbf{a}_R$ relative acc. in CS2 $\mathbf{a}_E$ Euler acc. $\boldsymbol{\epsilon} = \dot{\boldsymbol{\Omega}}$ $\mathbf{a}_C$ Coriolis acc. $\mathbf{a}_{CP}$ Centripetal acc.

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Modified Newton's method

Newton's method for the system specified in non-inertial frame of reference:



Newton's method is valid in CS1
(i.e. in inertial frame of reference)

EOM of rotating system using
CS2 (noninertial CS)

$$\mathbf{F}(t) = m\ddot{\mathbf{r}}$$

$$\mathbf{F}(t) - m\mathbf{a}_2 - m\boldsymbol{\varepsilon} \times \mathbf{r} - m2\boldsymbol{\Omega} \times \mathbf{v}_2 - m\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}) = 0$$

$$\mathbf{F}(t) + \mathbf{F}_2 + \mathbf{F}_E + \mathbf{F}_{Cor} + \mathbf{F}_{CF} = 0$$

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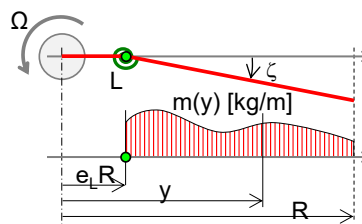
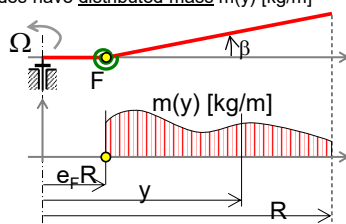
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Modelling blades as 1 DOF systems



Blades in rotors can be modelled as 1 DOF systems representing equivalent rotating flapping and lead-lagging rigid blades with these assumptions:

- small flapping angles β and lead-lag angles ζ
- flapping and lead-lag can be studied in isolation
- hingeless rotors have effective flapping hinges
- **hinges have non-dimensional flapping and lead-lag offsets** $e_F, e_L [-]$
- hinges can have equivalent torsional springs with stiffness $k_F, k_L [\text{N.m/rad}]$
- blades have distributed mass $m(y) [\text{kg/m}]$



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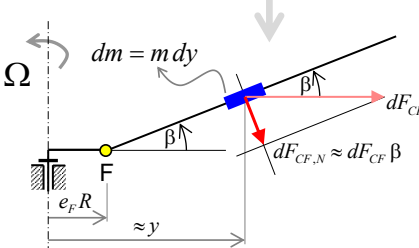
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Mechanics of flapping

We assume constant rotational speed Ω (provided by an ideal engine), gravity is not considered. We focus on a small blade segment and then add (integrate) contributions from *all* blade segments.

$\ddot{\mathbf{r}} = \underbrace{\mathbf{a}_2 + \cancel{\Omega \times \mathbf{r}} + \cancel{2\Omega \times \mathbf{v}_2}}_{\mathbf{a}_{2,n} + \mathbf{a}_{2,t}} + \underbrace{\Omega \times (\Omega \times \mathbf{r})}_{\mathbf{a}_{CP}}$

$\mathbf{F}(t) + \mathbf{F}_2 + \mathbf{F}_E + \mathbf{F}_{Cor} + \mathbf{F}_{CF} = 0$



Small flapping angles β : $|\mathbf{a}_{CP}| \approx \Omega^2 y$

Elementary centrifugal force dF_{CF} :

$$dF_{CF} = dm |\mathbf{a}_{CP}| = (m dy) \Omega^2 y$$

Normal component of dF_{CF} and dM_{CF} :

$$dF_{CF,N} = dF_{CF,N} \sin \beta \approx ((m dy) \Omega^2 y) \beta$$

$$\underline{dM_{CF} = dF_{CF,N} (y - e_F R)}$$

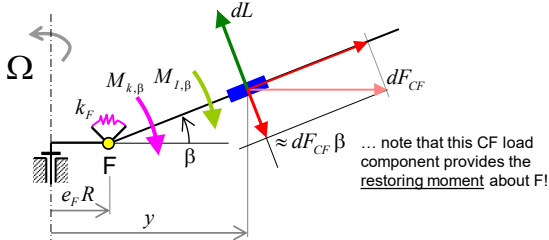
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Flapping EoM

Because we work with rigid blades, we will derive the overall (integral) loads acting on the rotating and flapping blade. These loads are calculated by *integrating* (or summing) the load contributions due to small blade elements across the span of the blade (from $e_F R$ to R).

- flapping blade
- no gravity
- constant Ω
- small β



Dynamic equilibrium: sum of all moment about F is equal to 0.

$$M_{I,\beta} + M_{CF,\beta} + M_{k,\beta} - M_{A,\beta} = 0$$

- $M_{I,\beta}$... inertial moment for blade flapping about F (Newton's method) (icing!)
- $M_{CF,\beta}$... moment about F due to centrifugal loads
- $M_{k,\beta}$... restoring moment due to torsional spring located in F
- $M_{A,\beta}$... moment about F due to external (aerodynamic) loads

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Flapping EoM

Local inertial moment: $M_{I,\beta} = I_{B,F} \ddot{\beta}$ $I_{B,F} = \int_{e_F R}^R (dist.)^2 dm = \int_{e_F R}^R (y - e_F R)^2 m dy \approx m R^3 (1/3 - e_F)$

Restoring moment due to torsional spring: $M_{k,\beta} = k_F \beta$

Moment due to aerodynamic loads: $M_{A,\beta} = \int_{e_F R}^R (y - e_F R) dL$

Moment due to centrifugal loads: $M_{CF,\beta} = \int_{e_F R}^R dM_{CF} \approx m \Omega^2 \beta R^3 \left(\frac{1}{3} - \frac{e_F}{2} \right)$

Flapping equation of motion:

$$\underbrace{(m R^3 (1/3 - e_F))}_{\text{inertial term}} \ddot{\beta} + \underbrace{(k_F + m \Omega^2 R^3 (1/3 - e_F/2))}_{\text{stiffness term}} \beta = \underbrace{M_{A,\beta}}_{\text{Aero loads}}$$

$$\ddot{\beta} + \underbrace{\frac{m \Omega^2 R^3 (1/3 - e_F/2) + k_F}{m R^3 (1/3 - e_F)}}_{\text{natural frequency of flapping motion}} \beta = \frac{1}{I_{B,F}} M_{A,\beta}$$

$$\ddot{\beta} + \omega_{F1}^2 \beta = M_{A,\beta} / I_{B,F}$$

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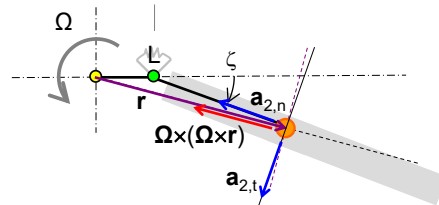
Mechanics of blade lead-lag

Lead-lag kinematics:

$$\ddot{\mathbf{r}} = \mathbf{a}_2 + \cancel{\mathbf{a}_2 \times \mathbf{r}} + \cancel{2\boldsymbol{\Omega} \times \dot{\mathbf{r}}_2} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r})$$

$$\mathbf{a}_{2,n} + \mathbf{a}_{2,t}$$

$$\mathbf{F}(t) + \mathbf{F}_2 + \cancel{\mathbf{F}_E} + \cancel{\mathbf{F}_{Cor}} + \mathbf{F}_{CF} = 0$$



Contribution to the restoring moment due to centrifugal force dF_{CF} :

$$dM_{CF} = dF_{CF,N} (y - e_L R) = (m dy) \Omega^2 (e_L R) \sin \zeta (y - e_L R)$$

$$dF_{CF} = (m dy) \Omega^2 r$$

dF_{CF} and $dF_{CF,N}$ can be derived using the geometry of the lead-lagging blade. Derivation not included.

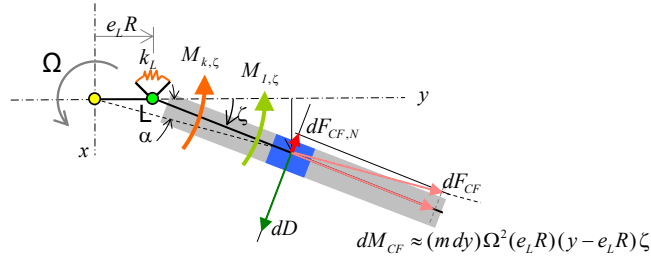
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Lead-lag EoM

Similar to the flapping blade analysis, we identify all loads (moments and forces) acting on the blade element that contribute to the equilibrium condition:

- lead-lag motion
- no gravity
- constant Ω
- small ζ



Dynamic equilibrium: sum of all moments about L is equal to 0.

$$M_{I,\zeta} + M_{CF,\zeta} + M_{k,\zeta} - M_{A,\zeta} = 0$$

- $M_{I,\zeta}$... inertial moment for blade moving about L (Newton's method)
- $M_{CF,\zeta}$... moment about L due to centrifugal loads (different from flapping!)
- $M_{k,\zeta}$... restoring moment due to torsional spring in L
- $M_{A,\zeta}$... moment about L due to external (aerodynamic) loads (e.g. drag)

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Lead-lag EoM

Local inertial moment: $M_{I,\zeta} = I_{B,L} \ddot{\zeta}$ $I_{B,L} = \int_{e_L R}^R (dist.)^2 dm = \int_{e_L R}^R (y - e_L R)^2 m dy \approx m R^3 (1/3 - e_L)$

Restoring moment due to torsional spring: $M_{k,\zeta} = k_L \zeta$

Moment due to aerodynamic loads: $M_{A,\zeta} = \int_{e_L R}^R (y - e_L R) dD$

Moment due to centrifugal loads: $M_{CF,\zeta} = \int_{e_L R}^R dM_{CF} \approx m \Omega^2 \zeta e_L R^3 \left(\frac{1}{2} - e_L \right)$

Equation of motion:

$$\underbrace{(m R^3 (1/3 - e_L))}_{\text{inertial term}} \ddot{\zeta} + \underbrace{(m \Omega^2 e_L R^3 (1/2 - e_L) + k_L)}_{\text{stiffness term}} \zeta = \underbrace{M_{A,\zeta}}_{\text{Aero loads}}$$

$$\ddot{\zeta} + \underbrace{\frac{m \Omega^2 e_L R^3 (1/2 - e_L) + k_L}{m R^3 (1/3 - e_L)}}_{\text{natural frequency of lead-lag motion}} \zeta = \frac{1}{I_{B,L}} M_{A,\zeta} \quad \ddot{\zeta} + \omega_{L1}^2 \zeta = M_{A,\zeta} / I_{B,L}$$

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Rigid blade dynamics

Flapping blade about hinge F:

$$\ddot{\beta} + \frac{m\Omega^2 R^3 (1/3 - e_F/2) + k_F}{mR^3 (1/3 - e_F)} \beta = \frac{M_{A\beta}}{I_{B,F}}$$

Lead-lag motion about hinge L:

$$\ddot{\zeta} + \frac{m\Omega^2 e_L R^3 (1/2 - e_L) + k_L}{mR^3 (1/3 - e_L)} \zeta = \frac{M_{A,\zeta}}{I_{B,L}}$$

ω_{F1} and ω_{L1} depend on (design and operational parameters):
 $R, \Omega, e_F, e_L, m(y), k_F, k_L$

ω_{F1} will be also denoted as F1

ω_{L1} will be also denoted as L1

Natural frequency of flapping motion:

$$\omega_{F1} = \Omega \left(1 + \frac{3e_F}{2(1-3e_F)} + \frac{k_F}{I_{B,F}\Omega^2} \right)^{1/2}$$

Natural frequency of lead-lag motion:

$$\omega_{L1} = \Omega \left(e_L \frac{(1/2 - e_L)}{(1/3 - e_L)} + \frac{k_L}{I_{B,L}\Omega^2} \right)^{1/2}$$

Dimensionless frequency F1:

$$\lambda_{F1} = \left(1 + \frac{3e_F}{2(1-3e_F)} + \frac{k_F}{I_{B,F}\Omega^2} \right)^{1/2}$$

Dimensionless frequency L1:

$$\lambda_{L1} = \left(e_L \frac{(1/2 - e_L)}{(1/3 - e_L)} + \frac{k_L}{I_{B,L}\Omega^2} \right)^{1/2}$$

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Rigid blade natural frequencies

• Flapping motion

$$\lambda_{F1} = \left(1 + \frac{3e_F}{2(1-3e_F)} + \frac{k_F}{I_{B,F}\Omega^2} \right)^{1/2}$$

- $\omega_{F1} = \lambda_{F1}\Omega$ [rad/s]
- origins of stiffness:
 - structural stiffness, k_F
 - CF stiffening, $\Omega \neq 0$
- special cases:
 - $\Omega \neq 0, e_F > 0, k_F > 0 \Rightarrow \lambda_{F1} > 1$
 - $\Omega \neq 0, e_F = 0, k_F = 0 \Rightarrow \lambda_{F1} = 1$
 - $\Omega = 0, k_F = 0 \Rightarrow \lambda_{F1} = 0$
- typical values:
 - $\lambda_{F1} \approx 1.06$ (articulated),
 - $\lambda_{F1} > 1.06$ (hingeless),
 - $e_F \approx 0.04$

• Lead-lag motion

$$\lambda_{L1} = \left(e_L \frac{(1/2 - e_L)}{(1/3 - e_L)} + \frac{k_L}{I_{B,L}\Omega^2} \right)^{1/2}$$

- $\omega_{L1} = \lambda_{L1}\Omega$ [rad/s]
- origins of stiffness:
 - structural stiffness, k_L !
 - CF stiffening, $\Omega \neq 0$
- special cases:
 - $\Omega \neq 0, e_L > 0, k_L > 0 \Rightarrow \lambda_{L1} > 0$
 - $\Omega \neq 0, e_L = 0, k_L = 0 \Rightarrow \lambda_{L1} = 0$
 - $\Omega = 0, k_L = 0 \Rightarrow \lambda_{L1} = 0$
- typical values:
 - $\lambda_{L1} \approx 0.3$ (articulated),
 - $\lambda_{L1} \approx 0.6$ (hingeless),
 - $e_L \approx 0.05$

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Blade frequency diagrams

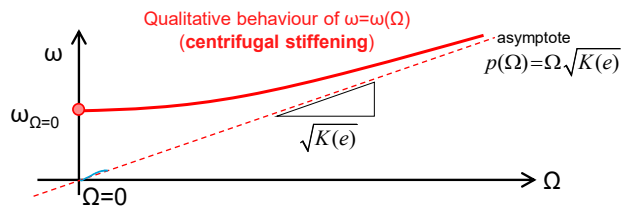
The important tool in *dynamics of rotors* are frequency diagrams (also called Southwell, Campbell, Fan, etc. diagrams). They show the effect of *rotation* across the range of speeds allowing thus to make the design decisions! They provide visual representation of the F1 and L1 change with increasing rotor speed Ω .

Consider the special cases of F1 and L1 to appreciate the main qualitative properties of the *natural frequency loci* (parametric paths $\omega=\omega(\Omega)$):

$$\omega_{F1} = \Omega \left(1 + \frac{3e_F}{2(1-3e_F)} + \frac{k_F}{I_{BF}\Omega^2} \right)^{1/2} \quad \omega_{L1} = \Omega \left(e_L \frac{(1/2 - e_L)}{(1/3 - e_L)} + \frac{k_L}{I_{BL}\Omega^2} \right)^{1/2}$$

$$\omega_{F1} = \sqrt{k_F/I_{BF}} \quad \omega_{F1} \approx \Omega(1 + 3e_F/(2(1-3e_F)))^{1/2} \quad \omega_{L1} = \sqrt{k_L/I_{BL}} \quad \omega_{L1} \approx \Omega(e_L(1/2 - e_L)/(1/3 - e_L))^{1/2}$$

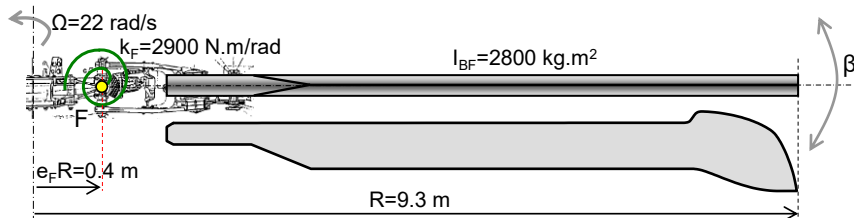
$\Omega=0$ (no rotation) $\Omega \rightarrow \infty$ (asymptote) $\Omega=0$ (no rotation) $\Omega \rightarrow \infty$ (asymptote)



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Example: AW101 F1 frequency



Formula for the natural frequency of flapping from the previously derived EOM:

$$\omega_{F1}^2 = \frac{m\Omega^2 R^3 (1/3 - e_F/2) + k_F}{mR^3 (1/3 - e_F)} = \Omega^2 \left(\frac{(1/3 - e_F/2)}{(1/3 - e_F)} + \frac{k_F}{I_{BF}\Omega^2} \right)$$

Non-dimensional flapping hinge offset:

$$e_F R = 0.4 \text{ m} \Rightarrow e_F = (0.4 \text{ m}) / (9.3 \text{ m}) = 0.043 \text{ (4.3\%)}$$

Natural frequency of flapping:

$$\omega_{F1} = \Omega \left(\frac{(1/3 - e_F/2)}{(1/3 - e_F)} + \frac{k_F}{I_{BF}\Omega^2} \right)^{1/2} = (22 \text{ rad/s}) \left(\frac{(1/3 - 0.043/2)}{1/3 - 0.043} + \frac{2900 \text{ Nm/rad}}{(2800 \text{ kg.m}^2)(22 \text{ rad/s})^2} \right) = 22.823 \text{ rad/s}$$

Non-dimensional (natural) frequency of flapping: $\lambda_{F1} = \frac{\omega_{F1}}{\Omega} = 1.037$

(λ_{F1} reported in available literature is 1.034, i.e. $\sim 0.3\%$ difference) (λ_{F1} for Lynx BERP is ~ 1.11)

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Summary

- Understand the significance, origin and applications of the *modified* Newton's method.
- CF loads act as *restoring* loads that:
 - restore original equilibrium position (i.e. stabilize),
 - provide positive stiffness,
 - depend on Ω .
- Flapping (lead-lagging) rigid blade is modelled as 1-DOF oscillator
 - understand the F1 and L1 frequency diagram.
- F1, L1 parameter influences and general properties:
 - rotor radius, speed, hinge offset, stiffness, mass distribution,
 - $(\omega_{F1}/\Omega) \geq 1$ (F hinge absorbs the effect of the “in-flight resonance”),
 - $(\omega_{L1}/\Omega) \geq 0$ (L motion is stabilized using added damping).

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